SMOOTH COUNTEREXAMPLES TO THE C^∞ LOEWNER AND CARATHÉODORY UMBILIC-INDEX CONJECTURES

ARGUS OPEN RESEARCH PROJECT

Abstract. Problem 8.2 in Ghomi's problem list on geometry of curves and surfaces asks to show that the index of any isolated singularity of a principal line field on a surface is at most 1 (Loewner's conjecture). While Hamburger proved the real-analytic case, we document and verify the C^∞ counterexample family g_k of Alpöge (2026) and its Lean 4 formalization, which constructs a C^∞ surface germ at the origin with an isolated umbilic of index $1 + k/2 = 3/2 > 1$ at $k = 1$ (and a smooth convex body with a single umbilic of index 2 at $k = 2$).

1. Summary of the Verified Formal Theorem

Theorem 1.1 (Alpöge, 2026; Lean 4 verified)**.** *In the C^∞ category:*

(1) *For every integer $k \geq 1$, there exists a C^∞ function $g_k : \mathbb{R}^2 \to \mathbb{R}$ whose trace-free Hessian $\left(\partial_{xx} g_k - \partial_{yy} g_k,\ 2\partial_{xy} g_k\right)$ has an isolated zero at the origin with winding number $2 + k$, corresponding to a principal-line umbilic index of $1 + k/2 > 1$.*

(2) *Taking $k = 1$ gives an isolated C^∞ umbilic of index $3/2 > 1$, refuting the C^∞ Loewner conjecture.*

(3) *Taking $k = 2$ and extending to S^2 yields the support function of a C^∞ strictly convex body in $\mathbb{R}^3$ with a unique umbilic point (of index 2), refuting the C^∞ Carathéodory conjecture.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 7000029 in the domain of differential geometry and smooth foliations follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *formal artifact refutation* deciding the scope: Alpoge's C^{ }infty surface family g_k with isolated umbilic of index 1+k/2 = 3/2 > 1 at k=1 (Lean 4 verified across all 4 modules).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- real-analytic surfaces (Hamburger's theorem).

References

[1] Primary problem list and source literature for Result 7000029 (differential geometry and smooth foliations), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: August 27, 2026.

2020 *Mathematics Subject Classification.* Primary 53A05; Secondary 37E35, 58K45.